

NETWORKS AND COMPLEXITY

Exercise Sheet 19: The Most Important Node

*This is an exercise sheet from the forthcoming book Networks and Complexity.
Find more exercises and solutions at <https://github.com/NC-Book/NCB>*

Ex 19.1: Eigenvectors [Lvl. 1]

For the Jacobian below, compute the eigenvalues λ_n , left eigenvectors $\mathbf{w}_n^\dagger$ and right eigenvectors $\mathbf{v}_n$, such that $\mathbf{w}_n^\dagger \mathbf{v}_m = \delta_{n,m}$. Then express the $\mathbf{x} = (0, 1)^\top$ as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

$$\mathbf{J} = \begin{pmatrix} -1 & 3 \\ 0 & -2 \end{pmatrix}$$

Ex 19.2: Observability and controllability [Lvl. 2]

For the Jacobian from the previous question compute the observability and controllability Gramians

$$\mathbf{P} = - \sum_{m,n} \frac{(\mathbf{w}_m \mathbf{w}_n^\dagger)(\mathbf{v}_m^\dagger \mathbf{v}_n)}{\lambda_n + \lambda_m^\dagger}, \quad \mathbf{Q} = - \sum_{m,n} \frac{(\mathbf{v}_m \mathbf{v}_n^\dagger)(\mathbf{w}_m^\dagger \mathbf{w}_n)}{\lambda_n + \lambda_m^\dagger}$$

and then compute the observability P_{ii} and controllability Q_{ii} of the two nodes.

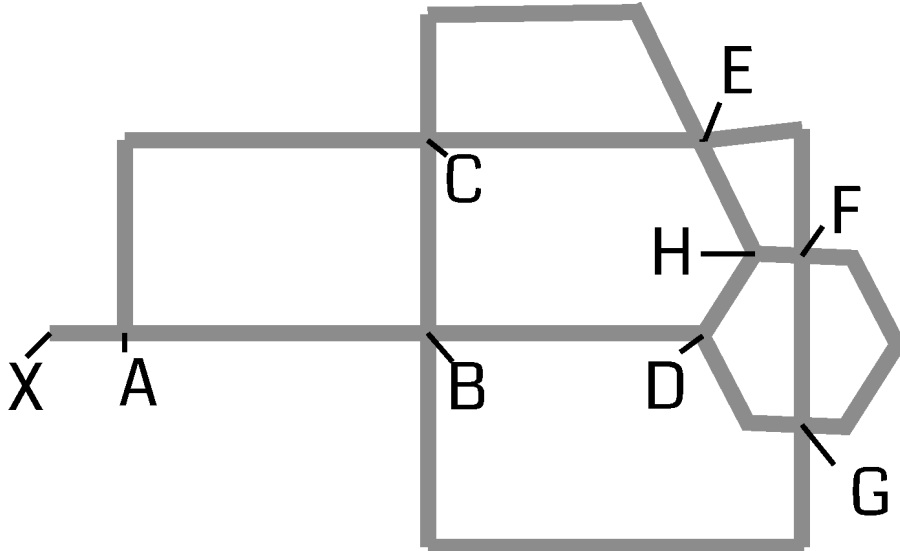
Ex 19.3: Linked pair [Lvl. 2]

Let's consider the linked pair (o-o) network in more detail.

- Write the adjacency matrix $\mathbf{A}$ and compute the eigenvalues λ_n , the right eigenvectors $\mathbf{v}_n$ and the left eigenvectors $\mathbf{w}_n^\dagger$.
- Consider the iteration $\mathbf{v}(i+1) = \mathbf{A}\mathbf{v}(i)$, starting with $\mathbf{v}(0) = (1, 2)^\top$. Find a closed-form solution for $\mathbf{v}(i)$.
- Now consider the shifted matrix $\mathbf{B} = \mathbf{A} + \mathbf{I}$, where $\mathbf{I}$ is an identity matrix of appropriate size. Again, find the eigenvalues and eigenvectors and study the iteration $\mathbf{v}(i+1) = \mathbf{B}\mathbf{v}(i)$, starting from $\mathbf{v}(0) = (1, 2)^\top$.

Ex 19.4: Traffic: Simple centralities [Lvl. 3]

Consider the following road network



- a) Compute the degree centrality of all nodes.
- b) Compute the closeness centrality of all nodes. (All links have length 1)
- c) Compute the betweenness centrality of all nodes.

Ex 19.5: Traffic: Spectral centrality [Lvl. 3]

Consider once again the road map from the previous exercise, but now compute the spectral centrality. (Mind the parallel links.)

Ex 19.6: Traffic: Summary [Lvl. 3]

In the past exercises we have computed several different types of centralities. For each of these state a context where the respective type of centrality of the road network would be relevant.

Ex 19.7: Divided attention [Lvl. 3]

Starting from the idea ‘important people have important friends’ derive another type of centrality. Proceed as in the case of spectral centrality, but this time nodes divide up their importance, such that for instance a neighbor of a node of degree four, only receives a quarter of the neighbor’s importance. Explore where this idea leads you. Consider the example of a 3-chain (o-o-o).

Ex 19.8: Epidemic Centrality [Lvl. 3]

Let’s find the most important node for a spreading process.

- a) Consider an SIS epidemic on a network described by an adjacency matrix $\mathbf{A}$. Explain why $x_i(t)$, the probability that node i is infected at time t , can be modeled by

$$\dot{x}_i = -rx_i + p(1 - x_i) \sum_j A_{ij}x_j$$

- b) Consider the limit where the disease is rare. In this limit, linearize the equations and determine the matrix $\mathbf{J}$ that allows you to write the system as $\dot{\mathbf{x}} = \mathbf{J}\mathbf{x}$
- c) The elements of the leading eigenvector of $\mathbf{J}$ now indicate who is most at risk in the epidemic. Explore how the largest eigenvalue λ and the corresponding eigenvector $\mathbf{v}$ are linked to the adjacency matrix $\mathbf{A}$.

Ex 19.9: Five star [Lvl. 4]

Consider a 5-star, a single central node of degree five that is connected to five nodes of degree 1. Start using the procedure from the chapter to compute the leading eigenvector. After a few steps use the insight gained to compute the eigenvector exactly. (This might be very hard or very easy)

Ex 19.10: Matrix perturbation theory [Lvl. 4]

Let’s suppose we make a small change to our Jacobian such that $\mathbf{J} = \mathbf{A} + p\mathbf{B}$, where $\mathbf{A}$ is the original Jacobian, $\mathbf{B}$ describes the change and p is a parameter that is initially zero. Let’s assume we know the eigenvalues λ_n and the left eigenvectors $\mathbf{w}_n^\dagger$ and the right eigenvectors $\mathbf{v}_n$ of $\mathbf{J}$ for $p = 0$. We now compute how the eigenvalues change as we increase p :

- a) Start with the (right) eigenvector equation for $\mathbf{J}$, substitute $\mathbf{J}$ and differentiate the whole equation with respect to p . Finally eliminate $\mathbf{A}$ by using the substitution backwards.
- b) You should now have one equation that contains the change of $\partial_p \lambda_n$ but also $\partial_p \mathbf{v}_n$, which we don’t want to compute. To fix this, multiply your equation with $\mathbf{w}_n^\dagger$ from the left, then find a way to eliminate $\partial_p \mathbf{v}_n$, and solve for $\partial_p \lambda_n$.
- c) Does the result give you an idea for another way in which we could rank the importance of nodes?

Ex 19.11: Press perturbation [Lvl. 4]

A dynamical system resides in a state $\mathbf{x}^*$. Your goal is to reduce the value of the first variable in the steady state, x_1^* . You can add a small constant term ε to one of the functions f on the right-hand side of your equation of motion. You recall the implicit function theorem

$$\frac{d\mathbf{x}^*}{dp} = -\mathbf{J}^{-1} \frac{\partial f}{\partial p}$$

where p is a parameter and $\mathbf{J}^{-1}$ is the inverse of the Jacobian $\mathbf{J}$ such that $\mathbf{J}\mathbf{J}^{-1} = \mathbf{J}^{-1}\mathbf{J} = \mathbf{I}$.

- a) Show that $\mathbf{J}^{-1}$ has the same left eigenvectors $\mathbf{w}_n^\dagger$ and right eigenvectors $\mathbf{v}_n$ as $\mathbf{J}$.
- b) Show that with orthonormal eigenvectors, the inverse of the Jacobian can be written as

$$\mathbf{J}^{-1} = \sum_n \frac{\mathbf{v}_n \mathbf{w}_n^\dagger}{\lambda_n}.$$

You can do this by showing that multiplying an arbitrary vector $\mathbf{a}$ from the right yields the same result on both sides of the equation.

- c) Now determine how the first coordinate of steady state x_1^* changes as we change when we add a small constant term ε to x_j 's equation of motion $f_j(\mathbf{x})$. Does this give you an idea for another centrality?

Ex 19.12: Multiple components [Lvl. 5]

Among the topological centralities (degree, closeness, betweenness and spectral centrality) only one extends naturally to networks with multiple components. Discuss the problems that occur, and try to find ways to extend all of these centralities gracefully to systems with multiple components.